

PART A — MODULE 1 QUESTIONS

MATH12-M1-001

Prompt:

What is the solution to $4 + (7x)/2 = 18$?

- A) 2
- B) 4
- C) 7
- D) 14

MATH12-M1-002

Prompt:

A recipe uses $\frac{3}{4}$ cup of honey for each batch of granola. How many cups of honey are needed for 8 batches of granola?

- A) 4
- B) 5
- C) 6
- D) 8

MATH12-M1-003

Prompt:

The table shows the number of club meetings attended by students in one month.

Meetings attended: 0, 1, 2, 3, 4

Number of students: 3, 5, 7, 4, 1

How many students attended fewer than 2 meetings?

- A) 5
- B) 8
- C) 12
- D) 15

MATH12-M1-004

Prompt:

A linear function f has $f(-2) = 10$ and $f(4) = -2$. What is $f(1)$?

- A) 2
- B) 4
- C) 6
- D) 8

MATH12-M1-005

Prompt:

A digital art program charges a monthly fee of \$18 plus \$4 for each image download. If d represents the number of image downloads in one month and the total cost is at most \$50, which inequality represents this situation?

- A) $18 + 4d \leq 50$
- B) $18 + 4d \geq 50$
- C) $4 + 18d \leq 50$
- D) $18 - 4d \leq 50$

MATH12-M1-006

Prompt:

What is the value of y in the solution to the system of equations?

$$2x - y = 1$$

$$x + y = 11$$

A) 4

B) 5

C) 7

D) 8

MATH12-M1-007

Prompt:

Which expression is equivalent to $(\frac{1}{2})(8n + 6) - 3n$?

A) $n + 3$

B) $n + 6$

C) $5n + 3$

D) $5n + 6$

MATH12-M1-008

Prompt:

Which expression is equivalent to $(m^2n^3)(m^4n)$?

A) m^6n^3

B) m^6n^4

C) m^8n^3

D) m^8n^4

MATH12-M1-009

Prompt:

A playlist contains 5 pop songs, 6 rock songs, and 7 jazz songs. If one song is chosen at random from the playlist, what is the probability that the song chosen is not a rock song?

A) $\frac{1}{3}$

B) $\frac{5}{9}$

C) $\frac{2}{3}$

D) $\frac{7}{9}$

MATH12-M1-010

Prompt:

A quadrilateral has angle measures of 90° , 85° , 110° , and x° . What is the value of x ?

A) 65

B) 70

C) 75

D) 85

MATH12-M1-011

Prompt:

In the xy -plane, the midpoint of segment AB is $(3, 5)$. Point A has coordinates $(1, 9)$. What are the coordinates of point B ?

A) $(2, 7)$

B) $(4, 1)$

C) $(5, 1)$

D) $(5, 9)$

MATH12-M1-012

Prompt:

What is the value of $3^2 + \sqrt{121}$?

- A) 14
- B) 18
- C) 20
- D) 22

MATH12-M1-013

Prompt:

If $f(x) = 2x^2 - x$, what is $f(-3)$?

- A) 15
- B) 18
- C) 21
- D) 24

MATH12-M1-014

Prompt:

The equation $x^2 - x - 20 = 0$ has two solutions. What is the sum of the two solutions?

- A) -20
- B) -1
- C) 1
- D) 20

MATH12-M1-015

Prompt:

In a survey of 80 students, 24 students said they ride a bike to school. What percent of the students surveyed said they ride a bike to school?

- A) 24%
- B) 30%
- C) 36%
- D) 40%

MATH12-M1-016

Prompt:

The table shows a linear relationship between x and y .

x : 2, 4, 6, 8

y : 7, 12, 17, 22

Based on the relationship shown, what is the value of y when $x = 10$?

- A) 25
- B) 27
- C) 29
- D) 32

MATH12-M1-017

Prompt:

A parallelogram has a base of 18 centimeters and a height of 7 centimeters. What is the area, in square centimeters, of the parallelogram?

- A) 25

- B) 63
- C) 126
- D) 252

MATH12-M1-018

Prompt:

A right triangle has a hypotenuse of length 26 and one leg of length 10. What is the length of the other leg?

- A) 16
- B) 20
- C) 24
- D) 36

MATH12-M1-019

Prompt:

The equation $H = 64 - 2t$ models the height H , in inches, of a plant t days after it was first measured. What does 64 represent in this model?

- A) The plant's height, in inches, when it was first measured
- B) The number of days the plant was measured
- C) The amount, in inches, that the plant's height changes each day
- D) The plant's height, in inches, after 64 days

MATH12-M1-020

Prompt:

The ratio of red buttons to blue buttons in a jar is 5:3. If there are 30 red buttons in the jar, how many blue buttons are in the jar?

- A) 12
- B) 15
- C) 18
- D) 24

MATH12-M1-021

Prompt:

The values in a data set are 2, 6, 8, 14, 16, and 18. What is the median of the data set?

- A) 8
- B) 10
- C) 11
- D) 14

MATH12-M1-022

Prompt:

Which ordered pair lies on the line represented by $5x - y = 12$?

- A) (2, -2)
- B) (3, 2)
- C) (4, 8)
- D) (5, 9)

PART B — MODULE 2 QUESTIONS

MATH12-M2-001

Prompt:

What is the solution to $2(3x + 5) - (x - 4) = 39$?

- A) 3
- B) 5
- C) 7
- D) 9

MATH12-M2-002

Prompt:

The system of equations has solution (x, y) .

$$2x + 3y = 27$$

$$4x - y = 19$$

What is the value of $2y - x$?

- A) 2
- B) 4
- C) 6
- D) 8

MATH12-M2-003

Prompt:

The function f is defined by $f(x) = 2x^2 - 16x + 35$. What is the minimum value of $f(x)$?

- A) -3
- B) 3
- C) 4
- D) 35

MATH12-M2-004

Prompt:

For $x \neq 3$, which expression is equivalent to $(x^2 - 7x + 12)/(x - 3)$?

- A) $x - 4$
- B) $x - 3$
- C) $x + 3$
- D) $x + 4$

MATH12-M2-005

Prompt:

A value is modeled by $1,200(0.92)^t$, where t is the number of years after the value was first measured. Which statement best describes the value according to this model?

- A) The value decreases by 8% each year.
- B) The value decreases by 92% each year.
- C) The value increases by 8% each year.
- D) The value increases by 92% each year.

MATH12-M2-006

Prompt:

After a 12% sales tax is added, the total cost of an item is \$84. What was the cost of the item before tax?

- A) \$72
- B) \$75
- C) \$78
- D) \$80

MATH12-M2-007

Prompt:

The table shows the number of students in two grades who visited a science museum.

Grade: 10, 11

Visited museum: 16, 18

Did not visit museum: 14, 12

If one student is selected at random from the students who visited the museum, what is the probability that the student is in Grade 11?

A) $\frac{8}{17}$

B) $\frac{9}{17}$

C) $\frac{3}{5}$

D) $\frac{18}{60}$

MATH12-M2-008

Prompt:

A rectangular sheet of cardboard is 14 inches long and 9 inches wide. A rectangular piece measuring 5 inches by 4 inches is cut from one corner. What is the area, in square inches, of the remaining cardboard?

A) 86

B) 96

C) 106

D) 126

MATH12-M2-009

Prompt:

In a right triangle, the hypotenuse has length 29 and the side adjacent to angle θ has length 20. What is $\sin \theta$?

A) $\frac{20}{29}$

B) $\frac{21}{29}$

C) $\frac{20}{21}$

D) $\frac{29}{21}$

MATH12-M2-010

Prompt:

Line r is represented by $2x - 5y = 10$. A second line is perpendicular to line r and passes through the point $(4, 3)$. What is the y -intercept of the second line?

A) -13

B) -7

C) 7

D) 13

MATH12-M2-011

Prompt:

The functions f and g are defined by $f(x) = x^2 + 2x$ and $g(x) = x - 5$. What is $f(g(8))$?

A) 9

B) 15

C) 24

D) 35

MATH12-M2-012

Prompt:

What is the solution to $\sqrt{x-2} + 4 = 10$?

- A) 34
- B) 36
- C) 38
- D) 42

MATH12-M2-013

Prompt:

A circle in the xy -plane has center $(-2, 5)$ and area 16π square units. Which equation represents the circle?

- A) $(x-2)^2 + (y+5)^2 = 16$
- B) $(x+2)^2 + (y-5)^2 = 16$
- C) $(x-2)^2 + (y+5)^2 = 4$
- D) $(x+2)^2 + (y-5)^2 = 4$

MATH12-M2-014

Prompt:

For what value of a does the equation $ax + 6 = 3x - 2$ have no solution?

- A) -2
- B) 0
- C) 3
- D) 6

MATH12-M2-015

Prompt:

The table shows values of a quadratic function q .

x : $-1, 0, 1, 2$
 $q(x)$: $8, 2, -2, -4$

Which equation could define q ?

- A) $q(x) = x^2 - 5x + 2$
- B) $q(x) = x^2 + 5x + 2$
- C) $q(x) = -x^2 - 5x + 2$
- D) $q(x) = -x^2 + 5x + 2$

MATH12-M2-016

Prompt:

Which expression is equivalent to $(14x^3z - 21x^2z^2)/(7x^2z)$, where $x \neq 0$ and $z \neq 0$?

- A) $2x - 3z$
- B) $2x - 21z$
- C) $7x - 3z$
- D) $7x - 21z$

MATH12-M2-017

Prompt:

A survey of 100 households found that 62 households subscribe to a streaming service, 48 households subscribe to cable, and 30 households subscribe to both. How many households subscribe to exactly one of these two services?

- A) 30
- B) 50
- C) 80
- D) 90

MATH12-M2-018

Prompt:

A material has a density of 0.8 gram per cubic centimeter. What is the mass, in grams, of 250 cubic centimeters of this material?

- A) 125
- B) 180
- C) 200
- D) 312.5

MATH12-M2-019

Prompt:

For all values of x , $5(x - 1)^2 + k = 5x^2 - 10x + 18$. What is the value of k ?

- A) 5
- B) 8
- C) 13
- D) 18

MATH12-M2-020

Prompt:

A data set has 7 values with a mean of 14. One value is added to the data set, and the mean of the 8 values becomes 16. What value was added?

- A) 22
- B) 26
- C) 30
- D) 34

MATH12-M2-021

Prompt:

At a cafe, muffins cost \$3 each and smoothies cost \$5 each. A customer buys a total of 24 muffins and smoothies for \$92. How many smoothies does the customer buy?

- A) 8
- B) 10
- C) 12
- D) 14

MATH12-M2-022

Prompt:

The graph of a quadratic function has x -intercepts at $x = 2$ and $x = 8$ and passes through the point $(0, 32)$. Which equation could define the function?

- A) $y = 2(x - 2)(x - 8)$
- B) $y = -2(x - 2)(x - 8)$
- C) $y = (x + 2)(x + 8)$
- D) $y = 2(x + 2)(x + 8)$

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH12-M1-001 — Correct: 4 — Explanation: Subtract 4 from both sides to get $(7x)/2 = 14$. Multiply both sides by 2 to get $7x = 28$, so $x = 4$.

MATH12-M1-002 — Correct: 6 — Explanation: Each batch uses $3/4$ cup of honey, so 8 batches use $8 \times 3/4 = 6$ cups.

MATH12-M1-003 — Correct: 8 — Explanation: Students who attended fewer than 2 meetings attended 0 or 1 meeting, so the total is $3 + 5 = 8$.

MATH12-M1-004 — Correct: 4 — Explanation: The rate of change is $(-2 - 10)/(4 - (-2)) = -12/6 = -2$. From $x = -2$ to $x = 1$, f decreases by 6, so $f(1) = 4$.

MATH12-M1-005 — Correct: $18 + 4d \leq 50$ — Explanation: The total cost is the monthly fee plus \$4 for each of d downloads, and this total is at most \$50.

MATH12-M1-006 — Correct: 7 — Explanation: Add the equations to get $3x = 12$, so $x = 4$. Then $x + y = 11$ gives $y = 7$.

MATH12-M1-007 — Correct: $n + 3$ — Explanation: Distribute $1/2$ to get $4n + 3 - 3n$, which simplifies to $n + 3$.

MATH12-M1-008 — Correct: m^6n^4 — Explanation: Multiply powers with the same base by adding exponents: $m^2m^4 = m^6$ and $n^3n = n^4$.

MATH12-M1-009 — Correct: $2/3$ — Explanation: There are 18 songs total and 6 are rock, so 12 are not rock. The probability is $12/18 = 2/3$.

MATH12-M1-010 — Correct: 75 — Explanation: The angle measures in a quadrilateral sum to 360° . Thus $x = 360 - 90 - 85 - 110 = 75$.

MATH12-M1-011 — Correct: $(5, 1)$ — Explanation: The midpoint coordinates are the averages of the endpoint coordinates. Since $(1 + x)/2 = 3$, $x = 5$; since $(9 + y)/2 = 5$, $y = 1$.

MATH12-M1-012 — Correct: 20 — Explanation: $3^2 = 9$ and $\sqrt{121} = 11$, so $9 + 11 = 20$.

MATH12-M1-013 — Correct: 21 — Explanation: Substitute -3 for x : $f(-3) = 2(-3)^2 - (-3) = 18 + 3 = 21$.

MATH12-M1-014 — Correct: 1 — Explanation: Factoring gives $x^2 - x - 20 = (x - 5)(x + 4)$, so the solutions are 5 and -4 . Their sum is 1.

MATH12-M1-015 — Correct: 30% — Explanation: The fraction is $24/80 = 0.30$, which is 30%.

MATH12-M1-016 — Correct: 27 — Explanation: Each time x increases by 2, y increases by 5. From $x = 8$ to $x = 10$, y increases from 22 to 27.

MATH12-M1-017 — Correct: 126 — Explanation: The area of a parallelogram is base $\times$ height, so the area is $18 \times 7 = 126$.

MATH12-M1-018 — Correct: 24 — Explanation: By the Pythagorean theorem, the other leg has length $\sqrt{(26^2 - 10^2)} = \sqrt{576} = 24$.

MATH12-M1-019 — Correct: The plant's height, in inches, when it was first measured — Explanation: In $H = 64 - 2t$, the constant 64 is the value of H when $t = 0$.

MATH12-M1-020 — Correct: 18 — Explanation: The ratio red:blue is 5:3. Since 30 red buttons represent 5 parts, each part is 6, so blue buttons total $3 \times 6 = 18$.

MATH12-M1-021 — Correct: 11 — Explanation: With 6 values, the median is the average of the 3rd and 4th values: $(8 + 14)/2 = 11$.

MATH12-M1-022 — Correct: $(4, 8)$ — Explanation: Substituting $(4, 8)$ into $5x - y = 12$ gives $5(4) - 8 = 12$, which is true.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH12-M2-001 — Correct: 5 — Explanation: Expand and combine like terms: $2(3x + 5) - (x - 4) = 6x + 10 - x + 4 = 5x + 14$. Then $5x + 14 = 39$, so $x = 5$.

MATH12-M2-002 — Correct: 4 — Explanation: From $4x - y = 19$, $y = 4x - 19$. Substitute into $2x + 3y = 27$ to get $2x + 3(4x - 19) = 27$. Then $14x = 84$, so $x = 6$ and $y = 5$. Thus $2y - x = 10 - 6 = 4$.

MATH12-M2-003 — Correct: 3 — Explanation: Complete the square: $2x^2 - 16x + 35 = 2(x - 4)^2 + 3$. Since the squared term is nonnegative, the minimum value is 3.

MATH12-M2-004 — Correct: $x - 4$ — Explanation: Factor the numerator: $x^2 - 7x + 12 = (x - 3)(x - 4)$. Since $x \neq 3$, the expression simplifies to $x - 4$.

MATH12-M2-005 — Correct: The value decreases by 8% each year. — Explanation: Multiplying by 0.92 each year means the value keeps 92% of its previous amount, so it decreases by 8% each year.

MATH12-M2-006 — Correct: \$75 — Explanation: If p is the pretax cost, then $1.12p = 84$. Dividing gives $p = 75$.

MATH12-M2-007 — Correct: $9/17$ — Explanation: A total of $16 + 18 = 34$ students visited the museum, and 18 of them are in Grade 11. The probability is $18/34 = 9/17$.

MATH12-M2-008 — Correct: 106 — Explanation: The original cardboard area is $14 \times 9 = 126$. The cutout area is $5 \times 4 = 20$. The remaining area is $126 - 20 = 106$.

MATH12-M2-009 — Correct: $21/29$ — Explanation: The opposite side has length $\sqrt{(29^2 - 20^2)} = \sqrt{441} = 21$. Therefore, $\sin \theta = \text{opposite/hypotenuse} = 21/29$.

MATH12-M2-010 — Correct: 13 — Explanation: The line $2x - 5y = 10$ has slope $2/5$. A perpendicular line has slope $-5/2$. Using point $(4, 3)$, $3 = (-5/2)(4) + b$, so $b = 13$.

MATH12-M2-011 — Correct: 15 — Explanation: First find $g(8) = 8 - 5 = 3$. Then $f(3) = 3^2 + 2(3) = 9 + 6 = 15$.

MATH12-M2-012 — Correct: 38 — Explanation: Subtract 4 from both sides to get $\sqrt{x - 2} = 6$. Squaring gives $x - 2 = 36$, so $x = 38$.

MATH12-M2-013 — Correct: $(x + 2)^2 + (y - 5)^2 = 16$ — Explanation: The area is 16π , so $\pi r^2 = 16\pi$ and $r^2 = 16$. With center $(-2, 5)$, the equation is $(x + 2)^2 + (y - 5)^2 = 16$.

MATH12-M2-014 — Correct: 3 — Explanation: If $a = 3$, the equation becomes $3x + 6 = 3x - 2$, which is false for all x . Therefore, the equation has no solution.

MATH12-M2-015 — Correct: $q(x) = x^2 - 5x + 2$ — Explanation: Substituting $x = -1, 0, 1$, and 2 into $x^2 - 5x + 2$ gives 8, 2, -2, and -4, matching the table.

MATH12-M2-016 — Correct: $2x - 3z$ — Explanation: Divide each term by $7x^2z$: $(14x^3z)/(7x^2z) - (21x^2z^2)/(7x^2z) = 2x - 3z$.

MATH12-M2-017 — Correct: 50 — Explanation: Streaming only is $62 - 30 = 32$, and cable only is $48 - 30 = 18$. Exactly one service is $32 + 18 = 50$ households.

MATH12-M2-018 — Correct: 200 — Explanation: Mass equals density times volume, so the mass is $0.8 \times 250 = 200$ grams.

MATH12-M2-019 — Correct: 13 — Explanation: Expand $5(x - 1)^2 + k$ to get $5x^2 - 10x + 5 + k$. Since this equals $5x^2 - 10x + 18$, $k = 13$.

MATH12-M2-020 — Correct: 30 — Explanation: The original sum is $7 \times 14 = 98$. The new sum is $8 \times 16 = 128$. The added value is $128 - 98 = 30$.

MATH12-M2-021 — Correct: 10 — Explanation: Let s be the number of smoothies. Then muffins are $24 - s$. The equation is $5s + 3(24 - s) = 92$. This gives $2s + 72 = 92$, so $s = 10$.

MATH12-M2-022 — Correct: $y = 2(x - 2)(x - 8)$ — Explanation: A quadratic with x -intercepts 2 and 8 has form $y = a(x - 2)(x - 8)$. Using point $(0, 32)$, $32 = a(-2)(-8) = 16a$, so $a = 2$.